HARYANA POLICE • STATE CRIME RECORDS BUREAU

SMART SURVEILLANCE USER GUIDE

Complete Operational Manual in Hinglish (ANPR + FRS)

Ye Haryana Police Surveillance System ka official operator manual hai. Isme frontend ke har ek button, switch, input field, modal, sliding drawer aur live AI detection pipeline ki step-by-step working saral **Hinglish** me explain ki gayi hai.

DOCUMENT ID:
HP-SURV-GUIDE-HINGLISH-V2.4

ACTIVE RELEASE:
Production v2.4 (Ready for Deployment)

CURRENTLY ACTIVE MODELS:
YOLO11n + Awiros PP-OCRv5 + SCRFD + ArcFace R34

FALLBACK ENGINE:
EasyOCR (Auto-Fallback Enabled)

WEB DASHBOARD PORT:
http://localhost:8000 • http://0.0.0.0:8000

LANGUAGE FORMAT:
Natural Hinglish (Roman Hindi + English Tech Terms)

© 2026 Haryana Police Department. Confidential • For Law Enforcement & Control Room Personnel Only.

CH 01 System Overview & Active AI Models

Hamara system Haryana ke highways, toll plazas, city entry gates aur sensitive locations par lagaane ke liye banaya gaya hai. Ye ek **Unified System** hai jo ek sath do kaam karta hai:

- **ANPR (Automatic Number Plate Recognition):** Gaadiyon (Car, Truck, Bus, Bike) ko detect karta hai, unki plate dhundhta hai aur plate number read karta hai.
- **FRS (Facial Recognition System):** Frame me maujood logo ke chehre detect karta hai aur police watchlists (Suspects/Wanted) ke sath match karta hai.

System me Currently Kon-Konse AI Models Use Ho Rahe Hain?

Project me strictly wahi models active hain jo humne configure aur verify kiye hain (koi fake ya extra model nahi hai):

AI Model Name	Model File Path	Framework / Size	Kaam (Exact Role in Pipeline)
YOLO11n (Vehicle)	`models/yolo/yolo11n.pt`	PyTorch / ~5.4 MB	Frame me cars, trucks, motorcycles aur buses ko detect karta hai aur **ByteTrack** se unhe track ID assign karta hai.
YOLO11_Plate	`models/yolo/yolo11_plate.pt`	PyTorch / ~5.5 MB	Gaadi ke andar number plate ke exact coordinates (tight bounding box) ko locate karta hai.
Awiros PP-OCRv5	`models/awiros_anpr/model.safetensors`	PaddlePaddle / 142.5 MB	**Hamara main Indian ANPR OCR engine.** Ye 37.3M parameter SVTR_HGNet CTC model hai jo Indian number plates ko high accuracy ke sath read karta hai.
EasyOCR Fallback	`src/ocr/ocr_engine.py`	PyTorch / Built-in	Agar kisi karan se Awiros model na mile ya load na ho, to ye system ko crash hone se bachata hai aur plate automatically read kar leta hai ("UNREADABLE" error nahi aane deta).
SCRFD 10G	`models/frs/scrfd_10g_bnkps.onnx`	ONNX Runtime / ~16.9 MB	Frame me se human faces ko detect karta hai aur 5 facial landmarks (dono aankhein, naak aur lips) nikaalta hai.
ArcFace ResNet-34	`models/frs/arcface_r34.onnx`	ONNX Runtime / 130.3 MB	Chehre ka 512-dimensional vector embedding nikaalta hai jise hum database me match karne ke liye use karte hain.

> 💡 **Dhyan Dein:** Heavy models (`model.safetensors` aur `arcface_r34.onnx`) private GitHub Releases (`v1.0-weights`) par securely host hain aur `python scripts/download_models.py` chala kar kisi bhi fresh machine par automatically download ho jaate hain.

CH 02 Navigation Sidebar & Main Menu

Dashboard ke left-hand side me persistent dark navigation bar diya gaya hai jisse operator bina page refresh kiye alag-alag modules me switch kar sakta hai:

Sidebar Button & ID	Button Type	Working (Button dabane se kya hota hai?)
`navDashboardBtn` **Live Surveillance Hub**	MAIN TAB	Is button ko dabane se aap **Live Monitoring Screen** par aa jaate hain jahan live camera feeds, 2x2 grid, aur video overlays dikhte hain.
`navMasterDbBtn` **Master DB Registry**	MAIN TAB	Is par click karne se **Criminal & Stolen Watchlists** khulti hain jahan chori ki gaadiyan aur wanted suspects ki entry ki jaati hai.
`navMasterTrailBtn` **Master Trail**	MAIN TAB	Is par click karne se **Master Trail** view khulta hai, jahan kisi gaadi ya shaqsiat ki timeline aur camera corridor tracking dikhti hai.
`navOpenVehiclesDrawerBtn` **Captured Vehicles**	SLIDE DRAWER	Screen ke right side se ek sliding drawer khul jaata hai jisme live cameras dwara capture ki gayi saari gaadiyan unke plate crop ke sath dikhti hain.
`navOpenFacesDrawerBtn` **Captured Faces**	SLIDE DRAWER	Screen ke right side se sliding drawer khulta hai jisme camera me aane wale har insaan ka face crop aur match score dikhta hai.

Global Status Badges (Header Bar)

🟢 Live Stream Beacon

Agar beacon green hai aur blink kar raha hai to iska matlab backend WebSocket aur camera feed live hai aur sub-50ms latency ke sath chal raha hai.

🚨 Live Hit Alarm

Jaise hi koi chori ki gaadi (Hotlist) ya wanted suspect camera ke samne aayega, header me red alarm bar flashing alert dikhane lagta hai.

CH 03 Live Surveillance Hub & HUD Overlays

Live Surveillance Hub par operator camera feed dekhta hai aur screen par dikhne wale boxes/text ko control karta hai.

1. View Mode Switcher (Screen Layout Buttons)

Button Name	HTML ID	Working & Output
1x1 Focus	`btnView1x1`	Is button ko dabane se grid band ho jaati hai aur **single selected camera** poori screen par maximize ho jaata hai taaki plate aur face saaf dikhein.
2x2 Grid	`btnView2x2`	Is button se screen **4 barabar hisson (CAM-01, CAM-02, CAM-03, CAM-04)** me divide ho jaati hai jisse ek sath 4 cameras monitor hote hain.
Fullscreen	`btnStreamFullscreen`	Video feed ko computer ki poori physical monitor screen par fullscreen mode me open karta hai.

2. Live HUD Preset Buttons (One-Click Layer Modes)

Agar operator ko baar-baar checkboxes on/off na karni ho to ye 4 preset buttons hain:

Preset Button	HTML ID	Working (Button dabate hi kya hota hai?)
Raw Feed	`btnPresetRaw`	Saare bounding boxes, boxes aur text band ho jaate hain. Screen par **100% clean video feed** dikhai deti hai bina kisi AI graphics ke.
ANPR Only	`btnPresetPlate`	Chehre aur insaan ke boxes band ho jaate hain; sirf **number plate ka box aur read kiya gaya plate number text** dikhta hai.
FRS Only	`btnPresetFrs`	Gaadiyon aur number plate ke boxes band ho jaate hain; sirf **human faces, 5 landmarks aur head count** dikhta hai.
All Layers	`btnPresetAll`	**Sabhi AI overlays ek sath chalu ho jaate hain:** Vehicle, ByteTrack ID, Plate box, Plate text, Person, Face, Landmarks aur Counter.

3. Individual Layer Checkboxes (Custom Filters)

Operator apni marzi ke hisab se kisi bhi visual element ko tick ya untick kar sakta hai:

Checkbox ID	Overlay Element	Working (Checkbox On/Off karne se kya badalega?)
`layerToggleVehicle`	Vehicle Box	Gaadi ke charon taraf green rectangular box dikhana ya chupana.
`layerToggleBytetrack`	ByteTrack ID	Gaadi ka unique continuous tracking number (e.g. `TRK-14`) dikhana.

Checkbox ID	Overlay Element	Working (Checkbox On/Off karne se kya badalega?)
`layerTogglePlate`	Plate Box	License plate ke charon taraf ka tight box toggle karna.
`layerToggleOcr`	OCR Text & Conf	Plate ke upar read kiya gaya text (e.g. `HR26AB1234`) aur confidence % dikhana.
`layerTogglePerson`	Person Box	Pedestrians (chalte huye logo) ke charon taraf cyan box toggle karna.
`layerToggleFace`	Face Box	Human face ke upar ka box. Normal face par amber, suspect par red dikhta hai.
`layerToggleLandmarks`	5 Landmarks	Chehre ke 5 alignment points (eyes, nose, mouth corners) draw karna.
`layerToggleCounter`	Live Tally Counter	Screen ke corner me live total vehicles aur active people count card dikhana.

CH 04 Connect Feed Modal & Tile Controls

Camera feed connect karne ke liye header me `btnOpenConnectModal` (**Connect Feed**) ya kisi tile par `+ Connect` par click karein.

1. Feed Types (Input Source Tabs)

1. 📹 **USB Webcam:** Local computer se jude webcam ko chalata hai (Index `0` laptop webcam, `1` external USB camera).
2. 🌐 **RTSP Stream:** Highway ya checkpost ke network IP camera ka RTSP link dalne ke liye (e.g., `rtsp://admin:pass@192.168.1.100:554/ch01`).
3. 🔗 **HTTP / HTTPS:** Web M-JPEG camera feeds ya live stream IP links ke liye.
4. 🎬 **Loop Demo:** Testing ya demo ke liye sample video files ko continuous loop me chalata hai.

2. Camera Configuration Fields & Buttons

Field / Button ID	Type	Working (Iska use kaise karein?)
Camera ID	Dropdown	Select karein ki ye camera grid ke kis tile me chalana hai: `CAM-01`, `CAM-02`, `CAM-03` ya `CAM-04`.
Camera Name / Location	Text Input	Camera ka naam likhein (e.g. "*Kundli Border Toll Lane 2*" ya "*Karnal Highway Checkpost*").
`btnDetectGps` 📍 **Auto-Detect GPS**	GPS BUTTON	**Auto-GPS Button:** Is button ko dabate hi browser location mangta hai aur system automatic Latitude aur Longitude fill kar deta hai.
`btnSubmitConnect` **Connect & Start Feed**	SUBMIT	Capture background thread launch karta hai aur live AI video stream start ho jaati hai.
`btnCancelConnectModal` **Cancel**	CANCEL	Modal ko band kar deta hai bina koi stream change kiye.
`btnDisconnectStream` **Disconnect**	DISCONNECT	Header bar me maujood ye button active running video stream ko stop karke release kar deta hai.

3. Per-Camera Tile Buttons (Hover Controls)

- **Focus Button (Maximize Icon):** Us specific camera ko turant 1x1 full view me le aata hai.
- **Disconnect Button (Cross Icon):** Sirf us specific channel ki streaming band karta hai.
- **Tile ANPR Checkbox (`tileToggleAnpr-CAM-XX`):** Agar system par CPU/GPU load kam karna ho to kisi camera par ANPR AI processing band karke sirf video chalu rakh sakte hain.

CH 05 Real-Time Sighting Drawers & Alarms

Camera ke samne se jaise hi koi gaadi ya vyakti nikalta hai, system automatic snapshot lekar sliding inspection drawer me entry add kar deta hai.

1. Captured Vehicles Drawer (`#vehiclesDrawer`)

Sidebar me `navOpenVehiclesDrawerBtn` dabane par right side se khulta hai:

Control / Button ID	Element Type	Working & Operation
`drawerVehFilterAll`	Filter Button	Live stream me detect huyi **sabhyi gaadiyon** ki list dikhata hai.
`drawerVehFilterHotlist`	Filter Button	Sirf wahi gaadiyan filter karta hai jo **Police FIR ya Chori (Stolen)** me wanted hain (Hotlist Hits).
`window.refreshDrawerVehicles`	Refresh Icon	SQLite database (`captured_db.sqlite`) se latest sightings fetch karke card list update karta hai.
`window.clearAllCapturedVehicles`	Clear Button	Confirmation mangne ke baad temporary live captured vehicles ki list ko clear kar deta hai.
`.side-drawer-close`	Close (×)	Sliding drawer ko wapas band karke live video screen ka poora view dikhata hai.

Sighting Detail Modal (`#captureDetailModal`):

Kisi bhi gaadi ke card par click karne par ye modal khulta hai jisme:

- **Full Scene Photo:** Camera ka poora frame jisme gaadi apni lane me dikhti hai.
- **Tight Plate Crop:** Number plate ka 4x magnified enhanced photo.
- **Plate Diagnostics:** Read kiya gaya number (e.g. `DL3CCY7011`), OCR confidence %, aur skew angle.
- **Hotlist Case Alert:** Agar gaadi stolen hai to FIR No, Police Station aur IO Contact details crimson red me flashing aati hain.

2. Captured Faces Drawer (`#facesDrawer`)

Sidebar me `navOpenFacesDrawerBtn` dabane par right side se khulta hai:

- `drawerFaceFilterAll` : Saare detect huye chehre dikhata hai.
- `drawerFaceFilterMatch` : Sirf wahi chehre dikhata hai jo police database ke suspects se **match** huye hain.
- `drawerFaceFilterSuspect` : Sirf **Wanted Criminals / FIR Accused** ko filter karta hai.
- `drawerFaceFilterVip` / `drawerFaceFilterStaff` : VIP ya Staff categorised chehre filter karta hai.
- **Face Sighting Modal (`#frsCaptureDetailModal`):** Chehre ka 112x112 aligned crop aur enrolled gallery photo ko side-by-side match percentage ke sath dikhata hai.

CH 06 Master DB Registry (Watchlists Management)

Master DB me police watchlists maintain ki jaati hain. Sidebar me `navMasterDbBtn` dabakar isme jate hain.

1. Vehicle DB Registry (`subtabVehicleDbBtn`)

Yahan stolen ya crime me involved gaadiyon ka record manage hota hai:

Control / Button ID	Type	Working & Police Use
`dbSearchInput`	Search Box	Plate number, FIR number, police station ya car color daal kar instant search karein.
All Records / FIR / Stolen	Filter Pills	Category wise gaadiyon ki table ko filter karne ke liye.
`btnOpenRegisterModal` **Register New Vehicle**	`MODAL TRIGGER`	Nayi wanted/stolen gaadi add karne ke liye registration form open karta hai.

Vehicle Registration Form Inputs (`#registerModal`):

- `regPlate` : Gaadi ka registration number (e.g. `HR26DA1234`).
- `regType` : Gaadi ki category select karein (Car, Motorcycle, Truck, Bus).
- `regColor` : Gaadi ka rang (White, Black, Silver, etc.).
- `regLocation` : Last known location ya owner ka address.
- `involvedInCase` : Radio button (Yes / No).
- `regOffenceType` : Apradh ki category (Theft, Robbery, Hit & Run, Kidnapping, Narcotics).
- `regFirNumber` : FIR Number (e.g. `FIR-142/2026`).
- `regPoliceStation` : Thana ka naam (e.g. *PS City Panipat*).
- `saveVehicleBtn` (**Save to Vehicle DB**): Record ko SQLite database (`data/vehicle_db.sqlite`) me permanently save karta hai.

2. Face DB Registry (`subtabFaceDbBtn`)

Yahan wanted suspects aur VIPs ke chehre enroll kiye jaate hain:

- `faceDbSearchInput` : Naam ya case ID se enrolled suspect ko search karein.
- `btnOpenEnrollModal` (**Enroll New Face**): Face enrollment form open karta hai.

Face Enrollment Form Inputs (`#enrollModal`):

- `enrollPersonName` : Suspect ka poora naam ya criminal alias.
- `enrollCategory` : Dropdown se category chunein (`SUSPECT` , `VIP` , `STAFF` , `SECURITY`).
- `enrollMetadata` : Case reference number ya criminal dossier ID.
- `enrollPhotoFile` : Suspect ki passport-size ya frontal photo upload karein. **System usi waqt ArcFace ResNet-34 model se 512-D vector feature embedding nikaal leta hai.**
- `submitEnrollBtn` (**Save to Face DB**): Record aur vector embedding ko `data/frs/face_db.sqlite` me save karta hai.

CH 07 Master Trail & Journey Corridor Tracking

Master Trail police investigation ka sabse zaroori tool hai. Ye kisi suspect gaadi ya vyakti ki **timeline** banata hai ki wo kis time par kis camera ke samne se guzra aur kis taraf ja raha hai.

Master Trail Controls

Control / Button ID	Type	Working (Investigation me kaise use karein?)
`subtabVehicleTrailBtn`	Subtab	Gaadiyon ke camera passage corridors dekhne ke liye.
`subtabFaceTrailBtn`	Subtab	Enrolled suspects ke camera passage corridors dekhne ke liye.
`trailSearchInput`	Search Box	Target plate number (e.g. `HR08AH3647`) daal kar corridor filter karein.
`btnApplyTrailFilters`	Filter Button	Date aur time range apply karta hai (e.g. pichle 24 ghante ka record).
`btnResetTrailFilters`	Reset Button	Filters hata kar complete history wapas load karta hai.
`btnToggleFaceTrailAuditTable`	Table Toggle	Visual timeline cards aur detailed spreadsheet (tabular audit log) ke beech switch karta hai.
`btnOpenTrailNarrativeModal`	AI Summary	**AI Route Summary:** Ek paragraph me police narrative banata hai ki gaadi kahan se ghusi, kis speed se gayi aur aakhiri baar kahan dekhi gayi.
`btnHeaderExportPdf`	Export PDF	**Download PDF Dossier:** Pure movement corridor ka print-ready official PDF report download kar deta hai jise case diary me lagaya ja sake.

> **Sample AI Route Summary Output:**
> *"Target Vehicle **HR-26-DA-1234** (Case FIR #204/24 PS Sadar) sabse pehle **09:14 AM** par Camera CAM-01 (Kundli Toll) par capture hua. Dakshin disha me lagbhag 60 km/h ki gati se chalte huye ye gaadi **09:38 AM** par Camera CAM-03 (Sector 14) par dekhi gayi, aur aakhiri baar **09:55 AM** par Camera CAM-02 (IFFCO Chowk Exit) par record huyi. Total recorded transit time: 41 minutes."*

CH 08 Unified Media Upload & Button Reference Table

Live camera ke alawa agar investigation officer ke paas mobile se khichi photo ya CCTV DVR ki video pen drive me ho, to use yahan test kiya ja sakta hai:

Unified Media Upload Hub Inputs

Input Control	HTML ID	Working & Supported Media
Unified Media Input	`unifiedMediaInput`	Photo ya Video dono accept karta hai aur dono engines (ANPR + FRS) ek sath run karta hai.
Video File Upload	`videoFileInput`	CCTV footage (MP4, AVI) frame-by-frame process hoti hai aur progress bar dikhta hai.
Image File Upload	`imageFileInput`	Single photo (JPG, PNG) dalne par instant dual-column analysis (Vehicle + Face) deta hai.
Close Inspection	`btnCloseInspection`	Forensic results band karke wapas live surveillance viewport par le aata hai.

Master Button Quick Reference (Sabhi Buttons Ka Index)

Button ID	Button Text	Kaam (Exact Single-Line Function)
`navDashboardBtn`	Live Hub	Live camera feeds aur live alerts screen open karta hai.
`navMasterDbBtn`	Master DB	Stolen gaadiyon aur wanted suspects ki registry open karta hai.
`navMasterTrailBtn`	Master Trail	Movement corridor aur timeline tracking screen open karta hai.
`btnView1x1`	1x1 Focus	Single camera ko screen par maximize karta hai.
`btnView2x2`	2x2 Grid	4 cameras ko ek sath 2x2 grid me arrange karta hai.
`btnOpenConnectModal`	Connect Feed	Naya RTSP/Webcam camera link karne ka dialog kholta hai.
`btnDisconnectStream`	Disconnect	Active video streams ko stop aur disconnect karta hai.
`btnPresetRaw`	Raw Feed	Saare AI boxes band karke 100% clean video dikhata hai.
`btnPresetPlate`	ANPR Only	Sirf plate box aur OCR plate number text dikhata hai.
`btnPresetFrs`	FRS Only	Sirf human face boxes aur facial landmarks dikhata hai.
`btnPresetAll`	All Layers	Sabhi AI visual overlays ko ek sath enable karta hai.
`btnDetectGps`	📍 Auto-Detect	Browser geolocation se station ke Latitude/Longitude auto-fill karta hai.

Button ID	Button Text	Kaam (Exact Single-Line Function)
`btnSubmitConnect`	Connect & Start	Camera stream background thread me start karta hai.
`btnOpenRegisterModal`	Register Vehicle	Chori ki gaadi ka FIR case record database me daalne ka modal kholta hai.
`saveVehicleBtn`	Save to Vehicle DB	Gaadi record ko `vehicle_db.sqlite` me commit karta hai.
`btnOpenEnrollModal`	Enroll New Face	Suspect ki photo aur dossier detail daalne ka modal kholta hai.
`submitEnrollBtn`	Save to Face DB	ArcFace se 512-D vector nikaal kar `face_db.sqlite` me save karta hai.
`btnHeaderExportPdf`	Export PDF	Investigative Movement Corridor report ka official PDF download karta hai.

END OF OPERATIONAL MANUAL (HINGLISH) • Haryana Police Surveillance Infrastructure Group